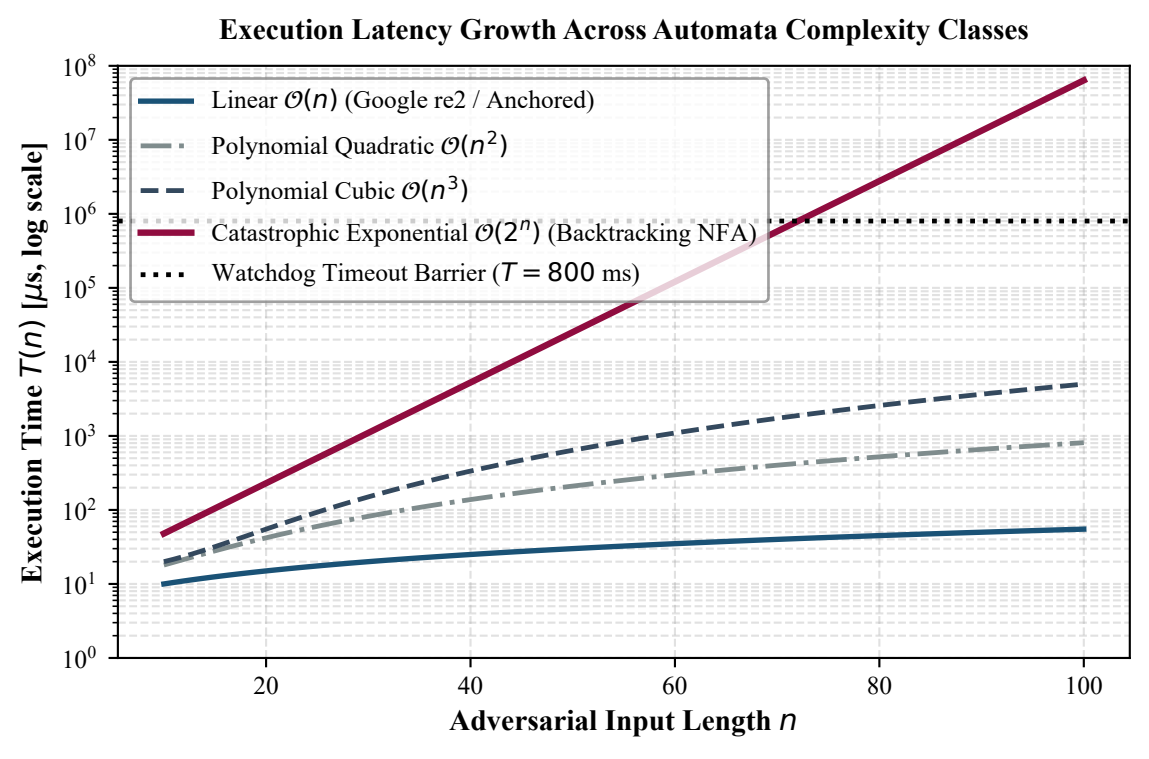

Execution Latency Growth Across Automata Complexity Classes
Linear $\mathcal{O}(n)$ (Google re2 / Anchored)
Polynomial Quadratic $\mathcal{O}(n^2)$
Polynomial Cubic $\mathcal{O}(n^3)$
Catastrophic Exponential $\mathcal{O}(2^n)$ (Backtracking NFA)
Watchdog Timeout Barrier ($T = 800$ ms)
Execution Time $T(n)$ [μs, log scale]
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
10^8
20
40
60
80
100
Adversarial Input Length n